

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

Our main goal is to study supernova light curves using data from the Mont4K CCD Camera, and previous measurement data to study their physical properties. Our observations will be conducted using the U (303-417nm), B (330-550nm), V (473-686nm), and R (550-900nm) filters on the Mont4K. Due to the uncertainty in the occurrences of supernovae, the targets will be finalized less than a week before observing. However, we can understand the potential features that can be analyzed using the obtained light curves. The type of supernova, the amount of dust around the supernova, and the mass of the exploding star can be interpreted from the photometry. The metallic composition can also be vaguely interpreted in the U-band at the outer layers, along with Hydrogen Alpha emission from circumstellar medium in the R-band.

Telescope Requested (0.5 points): Kuiper 61" on Mt. Bigelow

Instrument (0.5 points): Mont4K CCD Imager

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

The Moon would add brightness to the detector depending on its phase, which would skew the detected flux of the supernova. It would be preferred to not have the Moon rise at the time of observations, although there is some leeway for these observations. The minimum angular separation of the Moon from the target would be 30 degrees.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example “Object can be observed Feb-March.”

Our possible targets (These are just markers)

Supernova / Target	Right Ascension (RA)	Declination (Dec)	Calculated Peak (mag)	Object Type	Visibility
SN 2026wpd	01h 28m	+40° 13'	16.69 (367-561nm)	SN Ibn/Icn	Sep-Oct
SN 2026zyg	20h 35m	+51° 35'	18.05 (560-820nm)	SN II	Sep-Oct

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

As the observing target cannot be selected until closer to the observation date, we cannot calculate the exact signal-to-noise ratio (SNR) for our observations in advance. However, we estimate a minimum SNR of 5 for our observations. We also plan for 5 exposures per filter, and 180 seconds per exposure, resulting in 1 hour of observation per target. Including the time needed to change filters, read out the detector, acquire the target, and make other observing adjustments, we estimate that approximately 1.5 hours of observing time will be required.

The exposure time also needs to be considered in relation to the timescale over which the supernova fades. A supernova does not become significantly fainter within a few minutes; its brightness changes over a period of days to weeks as it evolves, and the radioactive material produced during the explosion decays. Therefore, a 3-minute exposure in each filter should provide enough time to collect sufficient photons while still giving us a measurement of the supernova at a specific point in its light curve.

The observations for this program will take place during the third weekend of October from the 16th to the 18th. However, since the data collected during these observations may not be sufficient, we will also consider incorporating data from other reliable sources.

Another important factor would be the airmass of the target. Airmass describes how much atmosphere the light from the target has to pass through before reaching the

telescope. We will therefore try to observe the target at an airmass below 1.5 whenever possible. An airmass of 2 or higher means that the light is passing through roughly twice as much atmosphere as it would at the zenith, so measurements at these elevations can be more affected by atmospheric extinction and changing sky conditions. Keeping the airmass low will help us obtain more consistent photometry and improve the reliability of the resulting light curve.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

When studying supernovas, the information depends heavily on whether the exploding star is a Type I or a Type II. Type I supernovas lack hydrogen in their light signatures while Type II supernovas contain plenty of hydrogen because they occur when a single, massive star runs out of fuel and its core collapses.

The light curve and photometry will help us determine how the brightness and energy of the supernova change over time. The number of peaks would inform us of the type of supernova. The width of the peak can be used to estimate the supernova's intrinsic color, which can then be compared with the observed color to calculate dust extinction. This will help understand how much dust is present around the supernova. The time it takes the supernova to reach its maximum brightness and the rate at which it becomes dimmer afterward can give us information about the explosion and the material produced during it. For example, in Type Ia supernovae, a broader and slower-declining light curve is generally associated with greater luminosity and a larger amount of Nickel-56 being produced, whose radioactive decay powers the later light curve.

These observations can also help us estimate the distance to the host galaxy. By comparing the observed brightness of the supernova with its expected intrinsic luminosity, especially Type Ia supernovae, we can estimate how far away it is. The shape of the light curve can also provide information about the material surrounding the exploding star. For example, Type IIP supernovae have a characteristic plateau in their light curves, which can tell us about the hydrogen-rich material involved in the explosion and the environment around the star. Overall, these observations will help us better understand how stars evolve and explode, how heavy elements are produced, and what remains after a supernova, such as a neutron star or black hole.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

Prof. Milne has offered to take data points on our target during his other nights at Kuiper 61", so those data points will act as our back up, and we will be able to continue with our project even when we can't take our data during our night.

In case Prof. Milne's observations turn out to be insufficient for the project, we will take data from this data archive: <https://lweb.cfa.harvard.edu/supernova/SNarchive.html>.